

Static Gesture Classification — Run Report

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Data paths

Training data path 1	/home/jestin/ThesisRepo/ML/NewTestData/Alan/Static · 35 trials
Training data path 2	/home/jestin/ThesisRepo/ML/NewTestData/Alex/Static · 35 trials
Training data path 3	/home/jestin/ThesisRepo/ML/NewTestData/Georgia/Static · 35 trials
Training data path 4	/home/jestin/ThesisRepo/ML/NewTestData/Henry/Static · 35 trials
Training data path 5	/home/jestin/ThesisRepo/ML/NewTestData/Jestin/Static · 36 trials
Training data path 6	/home/jestin/ThesisRepo/ML/NewTestData/Joselyn/Static · 30 trials
Training data path 7	/home/jestin/ThesisRepo/ML/NewTestData/Josh/Static · 35 trials
Training data path 8	/home/jestin/ThesisRepo/ML/NewTestData/Nat/Static · 37 trials
Total training paths	8 · 278 trials total
External test root	/home/jestin/ThesisRepo/ML/NewTestData/Marcus/Static
Report output dir	/home/jestin/ThesisRepo/ML/saved_reports

Sensor selection

Hands	left, right
Segments	palm_mid, thumb, index, middle, ring, pinky, wrist
Modalities	Flex
Sensor columns	20 channels
Classes	7: Double_Closed_Fist, Double_Nothing, Double_Okay, Double_Open_Palm, Double_Phone, Double_Pistol, Double_Spiderman

Preprocessing, features & split

Resample steps	90
Butterworth	on · cutoff 5.0 Hz · order 4 · fs 30.0 Hz
Normalisation	minmax
Feature mode	stats → feature dim 160
Test size · seed · CV folds	0.2 · random_state=42 · 10-fold stratified
Sample counts	train (post-aug)=444 · test=56

Augmentation (training only)

Enabled	yes
Copies per train trial	1
Jitter	off · sigma=0.01
Amplitude scale	on · range=(0.8, 1.2)
Time warp	off · range=(0.9, 1.1)

Classifiers — cross-validation & hold-out test

Algorithm	CV mean (10-fold)	CV std	Hold-out test acc
Random Forest	0.6062	0.0712	0.9286
Stacking (SVM+RF+KNN+LogReg)	0.7030	0.0610	0.8750
LDA	0.9639	0.0486	0.6786
Voting Soft (SVM+RF+KNN+LogReg)	0.7075	0.0482	0.6250
SVM (RBF)	0.7029	0.0605	0.4286
SVM (Linear)	0.6015	0.0545	0.4286
Logistic Regression	0.5493	0.0598	0.3750
KNN	0.3379	0.0303	0.2857

Per-class hold-out F1

Class	Random Forest	Stacking	LDA	Voting (soft)	SVM (RBF)	SVM (Linear)	Logistic Regression	KNN	Support
Double_Closed_Fist	1.000	1.000	0.500	0.824	0.778	0.706	0.533	0.240	9
Double_Nothing	0.923	0.833	0.667	0.556	0.471	0.333	0.333	0.444	7
Double_Okay	1.000	0.941	0.522	0.778	0.500	0.583	0.500	0.286	8
Double_Open_Palm	1.000	1.000	0.571	0.615	0.000	0.182	0.308	0.222	8
Double_Phone	0.889	0.842	0.857	0.667	0.632	0.667	0.588	0.273	8
Double_Pistol	0.875	0.800	0.800	0.364	0.000	0.000	0.154	0.333	8
Double_Spiderman	0.800	0.667	0.889	0.429	0.250	0.167	0.125	0.167	8
macro avg	0.927	0.869	0.687	0.604	0.376	0.377	0.363	0.281	56

External test (NewTestData)

Algorithm	External test acc
Random Forest	0.7714
Stacking (SVM+RF+KNN+LogReg)	0.7429

LDA	0.5714
Voting Soft (SVM+RF+KNN+LogReg)	0.5143
SVM (Linear)	0.4571
SVM (RBF)	0.4000
Logistic Regression	0.4000
KNN	0.1429

Per-class external accuracy

Class	Random Forest	Stacking	LDA	Voting (soft)	SVM (Linear)	SVM (RBF)	Logistic Regression	KNN	Support
Double_Closed_Fist	1.000	1.000	0.800	0.600	0.400	0.200	0.000	0.600	5
Double_Nothing	0.800	0.600	0.400	0.600	0.600	0.600	0.600	0.000	5
Double_Okay	1.000	1.000	0.400	0.800	0.600	0.600	0.400	0.000	5
Double_Open_Palm	0.000	0.000	0.400	0.000	0.400	0.000	0.200	0.000	5
Double_Phone	1.000	1.000	0.600	0.600	0.000	0.000	0.200	0.000	5
Double_Pistol	1.000	1.000	1.000	0.800	0.600	0.800	0.800	0.200	5
Double_Spiderman	0.600	0.600	0.400	0.200	0.600	0.600	0.600	0.200	5